Problem F
Question 1: Data Quality, Conflict Resolution, and Domain Mixture

Q1-internal validation, conflict diagnostics, and 17-domain mixture response

Final synchronized technical report

24 September 2026

Numerical source of truth. Every number, table and plotted quantity in this report is synchronized to the latest successful `python run_all.py` execution supplied by the team. Older report values are superseded.

Abstract

The latest Q1 pipeline retains 51,230 A1 records, explicitly audits semantic versus anchor-implied direction for all 22 indicators, fits an 11-feature anchor-aligned ridge quality proxy, applies it to the full A2/A3 extensions, and keeps conflict as a separate seven-family diagnostic. The mixture stage selects the quadratic centered-log-ratio family inside A4–A5 and locks $\lambda = 10$ before one A6–A7 blind evaluation. The blind test gives mean $R^2 = 0.8721$, median $R^2 = 0.8763$, mean MAE $= 0.2003$, and median Spearman $= 0.9515$. Cross-scale checks preserve ranking at 60M but degrade materially at 1B, 10B and 70B. The observed recipe used downstream has $\Delta_p^* = -0.530858$. Human raw-text reliability has not been completed and is reported as an unresolved evidence gap.

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1 Latest data flow

A1 contains 51,230 rows. Full extensions score 17,523 arXiv rows and 203,752 GitHub rows. Overlap consistency is exact to numerical precision: the 1,419 arXiv overlaps and 10,000 GitHub overlaps have Spearman 1.000000, with maximum absolute differences 0 and 4.163×10^{-17}. These are deterministic pipeline-consistency checks, not external validation.

2 Direction audit and quality score

The newest executable explicitly prints semantic and anchor-implied directions before fitting. Eleven indicators disagree in sign, including readability, reasoning, professionalism, fluency, the three DSIR components, advertising, unique-word fraction, top-3-gram fraction, and numerical-character fraction. This audit makes the orientation choices visible rather than implicit.

Table 1: Direction disagreements reported by the latest Q1 run.

Indicator	Semantic	Anchor-implied
modernbert_readability_mean	+1	-1
modernbert_reasoning_mean	+1	-1
modernbert_professionalism_mean	+1	-1
fluency_en_mean	+1	-1
dsir_books	+1	-1
dsir_wiki	+1	-1
dsir_math	+1	-1
ad_en_mean	-1	+1
rps_doc_frac_unique_words	+1	-1
rps_doc_frac_chars_top_3gram	-1	+1
rps_lines_numerical_chars_fraction	-1	+1

The final score uses 11 predictive features plus two anchors. The fitted anchor correlations are

$$\rho(Q_i, \text{qURATER}) = 0.5915, \qquad \rho(Q_i, \text{FineWebEdu}) = 0.4618.$$

Table 2: Normalized ridge weights in the final score.

Feature	w_{norm}
rps_doc_word_count	+0.2332
modernbert_readability_mean	+0.1321
fluency_en_mean	+0.1248
ad_en_mean	-0.1211
rps_doc_frac_no_alph_words	+0.0985
rps_doc_unigram_entropy	-0.0767
modernbert_professionalism_mean	+0.0613
dsir_composite	+0.0609
rps_doc_mean_word_length	-0.0399
rps_doc_frac_chars_top_2gram	+0.0261
rps_lines_uppercase_letter_fraction	-0.0253

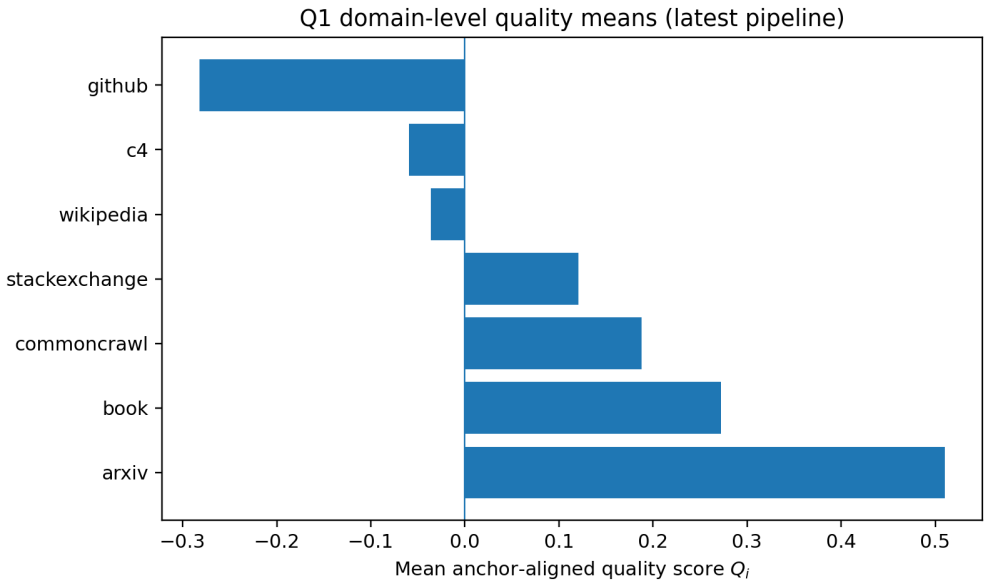


Figure 1: Latest A1 domain-level quality means.

Table 3: Full A1/A2/A3 quality summaries.

Dataset	Domain	n	Mean	SD	Median	q95
A1	arxiv	1419	0.510530	0.155076	0.530372	0.729369
A1	book	171	0.272550	0.176055	0.277937	0.574060
A1	c4	10000	-0.059230	0.379735	-0.032379	0.527051
A1	commoncrawl	9640	0.187785	0.267887	0.195312	0.613137
A1	github	10000	-0.282509	0.297936	-0.307585	0.224374
A1	stackexchange	10000	0.120888	0.256914	0.140751	0.520978
A1	wikipedia	10000	-0.036352	0.351017	-0.057506	0.564310
A2 full	arxiv	17523	0.507650	0.149299	0.525248	0.721663
A3 full	github	203752	-0.283080	0.300225	-0.312661	0.231518

3 Conflict as a separate diagnostic

Seven family-level oriented ranks produce the disagreement score

$$C_i = \frac{2}{G(G-1)} \sum_{g<h} |S_{ig} - S_{ih}|, \qquad G = 7.$$

Only 51,212 A1 rows have all 22 indicators. The thresholds are q85 = 0.292575, q90 = 0.313281, q95 = 0.343122. The q90 high-conflict rate is 10.0016% overall.

Table 4: Conflict summary.

Domain	n	Mean C	High-conflict	Mean confidence
arxiv	1419	0.281617	27.766%	0.992816
book	171	0.299760	36.257%	0.984948
c4	10000	0.252609	25.780%	0.989036
commoncrawl	9636	0.190237	4.618%	0.998510
github	10000	0.210661	8.450%	0.997279
stackexchange	10000	0.219798	5.950%	0.998676
wikipedia	9986	0.187134	2.033%	0.999521
ALL	51212	0.214469	10.002%	0.996446

The confidence-shrunk diagnostic has mean $Q_i = 0.0001$, mean resolved score 0.0006, and correlation 0.9999 with the original score. It is reported as a diagnostic; it is not fused back into the primary Q_i used downstream.

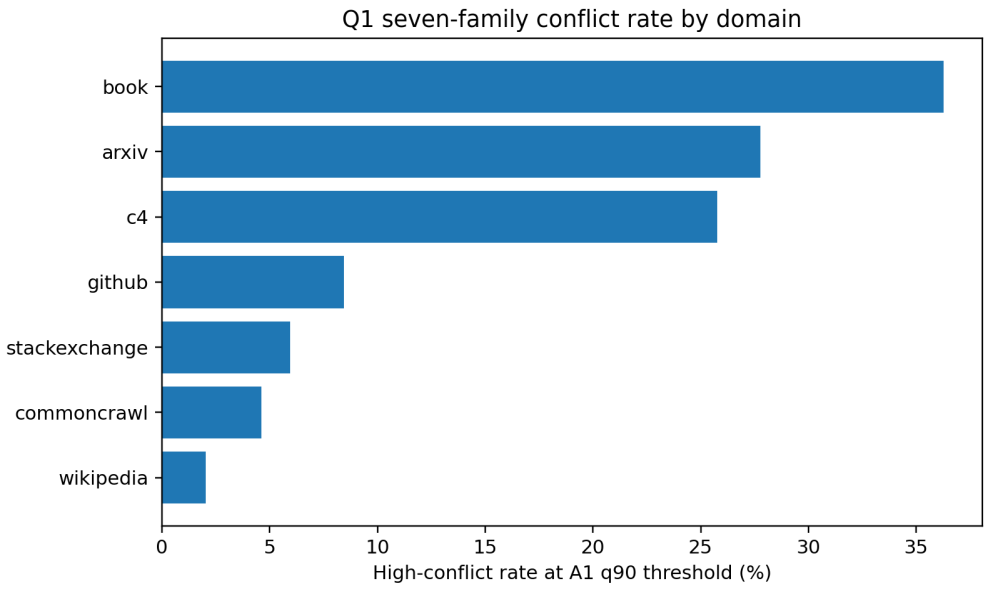


Figure 2: High-conflict rate by domain at q90.

4 Descriptive quality stratification

A two-component Gaussian mixture on Q_i gives a low component $(\mu, \sigma, \pi) = (-0.3456, 0.2515, 0.3567)$ and a high component $(0.1919, 0.2533, 0.6433)$. This is descriptive only and is not the requested 17-domain training-mixture model.

5 17-domain mixture-to-loss response

A4–A5 contains 512 recipes with 17 mixture shares and 13 loss outputs; A6–A7 contains 256 blind-test recipes. The final family is quadratic CLR with $\lambda = 10$, locked before the blind test. Fold-level outer MAEs are 0.2540, 0.2423, 0.2332, 0.2440, and 0.2301. The blind test is

$$\overline{R^2} = 0.8721, \quad \text{median } R^2 = 0.8763, \quad \text{MAE} = 0.2003, \quad \text{median } \rho = 0.9515.$$

Table 5: Additional cross-scale validation. Raw R^2 measures absolute transfer; centered R^2 removes a post-hoc per-domain mean shift.

Set	n	Raw R^2	Centered R^2	Spearman ρ
A8–A9 (60M)	256	-7.6782	+0.6389	0.9487
A10–A11 (1B)	64	-452.8249	-3.8015	0.8383
A12–A13 (10B est.)	63	-235.9140	-4.0876	0.6665
A14–A15 (70B est.)	63	-384.3578	-7.4907	0.5223

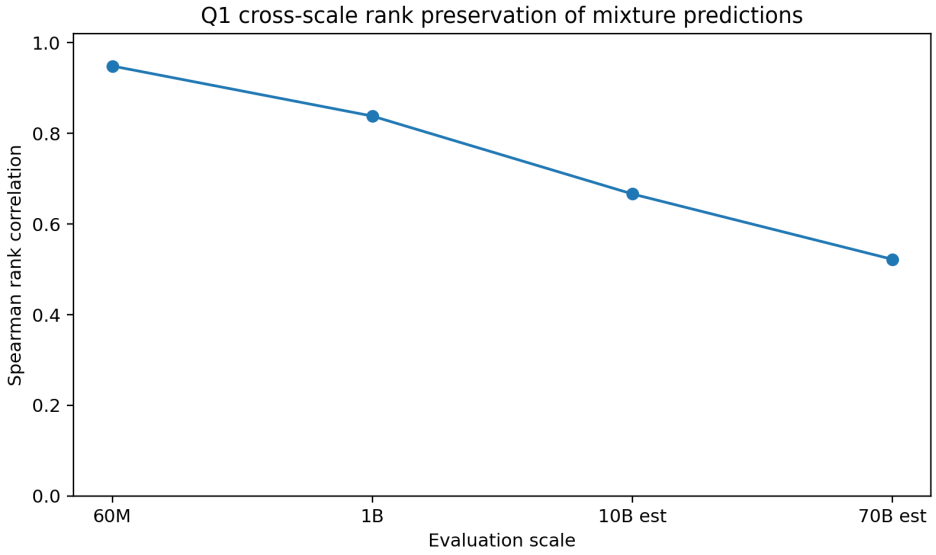


Figure 3: Rank preservation degrades as evaluation scale moves away from the 1M training scale.

6 Local mixture effects and downstream recipe

At the Aitchison center the macro loss is 5.174885. Selected one-percentage-point effects are shown below; negative is favorable.

Table 6: Selected local mixture effects.

Domain	ΔL	Domain	ΔL
dm_mathematics	-0.010528	enron_emails	+0.033498
ubuntu_irc	-0.008688	philpapers	+0.015461
stackexchange	-0.004000	nih_exporter	+0.008083
wikipedia_en	-0.001863	hackernews	+0.005907
github	-0.000887	europarl	+0.004246

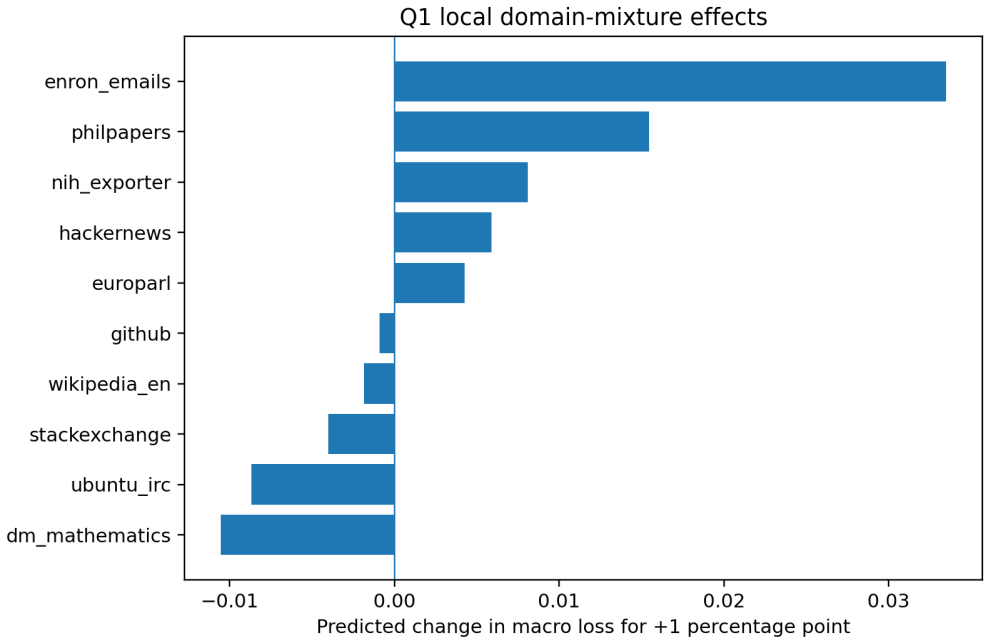


Figure 4: Latest local one-percentage-point macro mixture effects.

Selected favorable interaction residuals include wikipedia_en+enron_emails = −0.000545 and philpapers+pubmed_abstracts = −0.000441; the strongest adverse residual is philpapers+enron_emails = +0.005299.

The fixed observed recipe used in Q3 has

$$L(p^*) = 4.644027, \qquad L(p_0) = 5.174885, \qquad \boxed{\Delta_p^* = -0.530858}.$$

7 Known limitations - honest disclosure

Q1 limitations. (i) The human raw-text reliability panel has **not** been completed; no synthetic ratings are substituted. (ii) A1/A2/A3 overlaps are pipeline-consistency checks, not independent external validity. (iii) The quality score uses 11 predictive features derived from the 22 indicators; remaining signals still contribute to the conflict-family diagnostics. (iv) Conflict is a separate disagreement diagnostic and is not fused into the primary Q_i. (v) The A8–A15 absolute-transfer R^2 values become strongly negative beyond 60M, so cross-scale mixture claims should emphasize ranking and explicitly label estimated 10B/70B results.

8 Final Q1 answer

Q1 delivers a reproducible quality proxy, an auditable conflict score, and a simplex-aware 17-domain response surface. The strongest empirical result is the locked A6–A7 blind rank/fit performance; the strongest caution is that human reliability and large-scale absolute transfer remain unresolved.